

Improve productivity: DevOps trends often focus on improving automation and efficiency in the development process, which can result in improved productivity for your team.

Enhanced collaboration: Keeping up with DevOps trends can also lead to better collaboration between different teams and departments within an organisation.

Better quality software: Following DevOps trends can also help organisations deliver better quality software, through improved testing processes and faster release cycles.

Reduce costs: DevOps trends often emphasise reducing the time and costs associated with development and deployment, which can lead to significant cost savings for your organisation.

Current DevOps trends

DevOps is an essential methodology that helps organisations to improve their software development life cycle by aligning their development and operational teams. The DevOps culture is continuously evolving, and with more companies adopting this approach to improve their software delivery, we can see some significant DevOps trends emerging in the technology world. Let's take a look at some of the exciting and innovative trends that are driving DevOps forward.

Automation: Automating repetitive tasks such as server configuration, application deployment, testing, and monitoring is becoming increasingly popular to minimise human errors and speed up the delivery process.

Continuous delivery: Continuous delivery is aimed at ensuring that the software is always in a deployable state, and any changes made to the codebase can be released to production immediately.

Microservices: Rather than monolithic architecture, microservices use a collection of loosely coupled services to provide flexible, scalable, and independent development processes.

Cloud-native: Cloud-native emphasises deploying applications in the cloud instead of relying on traditional on-premises infrastructure.

Security-as-Code: Security is baked into the CI/CD pipeline so that automated testing detects vulnerabilities before they become threats.

DevSecOps: DevSecOps integrates security practices into the DevOps pipeline, so security becomes everyone's responsibility, rather than of a separate team.

Infrastructure as Code (IaC):

Infrastructure as Code is about defining and managing infrastructure using code to ensure reproducibility, consistency, and scalability.

GitOps: GitOps is a new approach to continuous delivery that uses Git as the single source of truth for managing infrastructure and applications.

Observability: Observability is the ability to monitor, understand, and troubleshoot complex systems in real-time.

Serverless computing: Serverless computing is an event-driven architecture that allows developers to deploy code without worrying about infrastructure management.

Automation and Infrastructure as Code (IaC)

Automation and Infrastructure as Code (IaC) are two closely related concepts that have revolutionised the way businesses build and manage their IT infrastructure. Automation refers to the use of tools and technologies to automate various IT operations, such as provisioning, configuration, testing, and deployment.

With automation, businesses can reduce the time and effort required to perform routine tasks, improve the accuracy and reliability of their IT operations, and free up IT teams to focus on more strategic initiatives. By automating various processes, businesses can also reduce costs and improve efficiency, enabling them

to deliver products and services faster and more reliably.

Infrastructure as Code, on the other hand, refers to the practice of using code to manage and provision infrastructure. With IaC, businesses can use software code to automate the deployment and configuration of their infrastructure, making it more scalable, repeatable, and reliable. IaC allows businesses to define their infrastructure as code, which can be easily versioned, tested, and reviewed, enabling them to make changes quickly and efficiently.

IaC has several benefits, including reducing the risk of human error by automating repetitive tasks, making it easier to manage complex infrastructure, and improving collaboration and communication between different teams. Additionally, IaC can help businesses achieve regulatory compliance, as they can easily track and audit changes to their infrastructure.

Security integration into DevOps

DevOps focuses on collaboration and communication between different departments such as software development, testing, and operations to ensure the seamless and continuous delivery of software products.

However, security has often been an afterthought in the DevOps process, with many organisations prioritising speed and agility over security. This approach can result in security vulnerabilities in the software products released, leading to data breaches, cyber-attacks, and loss of user trust, which can significantly impact an organisation's reputation and financial standing.

To address this challenge, security integration into DevOps is becoming a critical component of software development processes. This approach involves integrating security practices, tools, and workflows into the DevOps cycle to ensure that security is at the forefront of the software development process.